



SPARTA DSMC FAMILIARITY

Self-Study Master's Project

Abstract

This project will discuss the details of several example problems of increasing complexity performed with the SPARTA DSMC open source program from Sandia National Laboratory. There will be links to several videos that along with this report will allow the reader to replicate the results of these simulations.

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Project Purpose

The purpose of this project was to expand my knowledge base and to try to explore an subject outside of my comfort zone in engineering. I have taken several fluids classes and learned about the tools that are normally used to solve fluid dynamics problems both analytically and numerically. In several of my classes we have derived the Navier Stokes equations which accurately describe the physics of fluid dynamics in most cases. These equations are complex enough that they must be solved numerically for all but the most simplified cases. There are many commercial programs that do this well and I have taken a class where we learned to write our own programs for simple cases. One of the major assumptions made when using the Navier Stokes equations is that the fluid is modeled as a continuum. While this assumption is valid almost everywhere on earth there are conditions that can make this assumption invalid.

When the continuum assumption is no longer valid then we must change our perspective from the macroscopic model captured in the Navier Stokes equations to a microscopic model captured with the Boltzmann equation. Looking at a gas in this way is often called molecular gas dynamics or rarified gas dynamics. G. A. Bird has been one of the leaders in the development of ways to define and solve problems of this nature. His book titled *Molecular Gas Dynamics and the Direct Simulation of Gas Flows* [1] was the primary text that I used in my learning. One of the first sections discusses when the continuum assumption breaks down.

In chapter 1 of his book Bird discusses how the degree of rarefication is determined, and at what level the continuum assumption generally becomes invalid. The Knudsen number (Kn) is a ratio of the mean free path (λ) to the characteristic length (L). There is significant error in the Navier Stokes equations when this number is larger than 0.1 [1].

$$(Kn) = \frac{\lambda}{L}$$

By examining the ratio above we can see that as the mean free path increases and/or the characteristic length decreases that the Knudsen number will increase. Therefore, two of the primary areas of study and usage of the molecular model are in very low-density flows where there is a large mean free path between particles and micro channels related to surfaces with very small characteristic lengths. For my project I looked exclusively at low density flows. Which can occur at extremely high altitude here on earth, in an industrial

process that requires a vacuum or in the atmosphere of another planet which may be much less dense than the atmosphere of earth.

Once the Knudsen is above 0.1 the Boltzmann equation must be used to solve the dynamics of the gas particles which are still colliding with each other and surfaces. There is an upper limit to the Knudsen equation when one must switch to the collision less model which I will not be discussing in this project. Even as the Knudsen number gets larger the number of particles in gas volume is still exceptionally large which makes solving the Boltzmann equation for every particle impractical. This is what lead to the development of Direct Simulation Monte Carlo to simulate these flows.

G. A. Bird is credited with the development of Direct Simulation Monte Carlo (DSMC) in the 1970s and 1980's. The textbook that I used for this learning deeply discusses all the aspects of the method and how to deal with all the different physics that need to be considered. At the instruction of my advisor Dr. Gonzalez, I focused on the first 6 chapters of the book for basic understanding and then proceeded to chapter 12 to start working on problems.

Exploring the Original DSMC Codes

Bird wrote his original codes in Fortran, and they can be found on Git HUB at <https://github.com/schuberm/dsmc/tree/master/originalbird> and one of the more basic collision codes is in appendix D in Bird [1]. For this project I wanted to try and understand the basic mechanics of how his codes worked, so I converted his original programs for 1-D problems from Fortran to MATLAB. It was time intensive, but I felt like it helped me understand at least the sequence of how the simulations work. I used these codes later when I did my first comparisons with the SPARTA program. One note is that MATLAB is not the best option for these programs. They are computationally very heavy and C++ would have been a better choice. The MATLAB simulations took hours to run, while the program SPARTA that this project focused on using took minutes for the same simple 1-D simulations.

What is SPARTA?

One of the other references that I used throughout this project was the SPARTA instruction manual [3]. The manual is found at <https://sparta.github.io/doc/Manual.html>. This manual is very robust in the information it contains, however if you do not know what you are looking for in the manual it can be overwhelming. The manual [3] answers the questions about what SPARTA stands for and what SPARTA. “SPARTA stands for Stochastic Parallel Rarefied-gas Time-accurate Analyzer”, and “SPARTA is a Direct Simulation Monte Carlo code that models rarefied gases, using collision, chemistry, and boundary condition models.” Other useful information about SPARTA is that it was developed by Sandia National Laboratory and funded by the Department of Energy. The primary developers are Steve Plimpton, and Michael Gallis, and it is a completely open-source program, which is why I chose SPARTA for this project.

SPARTA, being an open-source program, comes with challenges. Open-source programs are generally not as user friendly as commercial programs that have large amounts of consumer revenue fueling the support and many of the ease-of-use features that are common with commercially available software. SPARTA is also only available for Linux systems, and the set up and use of the Linux operating system was new to me and provided an additional challenge. Then there is the challenge of how SPARTA works. There is no graphical user interface. All the simulation parameters must be contained in several text files that will be used by the program.

While there are examples in the SPARTA download there are very few examples of other problems online that show a step by step approach to solving the problem and the commands to use. My goal for this project is to provide detailed examples on how to set up and run SPARTA simulations. I will do a couple of 1-D examples from chapter 12 of Bird’s book [1] and then try and reproduce the SPARTA results for two problems that Angelos Kiothakis and Ioannis K. Nikolos produced in their journal article titled *Comprehensive Evaluation of the Massively Parallel Direct Simulation Monte Carlo Kernel “Stochastic Parallel Rarefied-Gas Time-Accurate Analyzer” in Rarefied Hypersonic Flows* [3]. For these examples, the required files will be included in the appendices, and there will be a video provided discussing the key components of these files. Comparisons of my results to the literature will be in the main written text of this report. So, while reading this project report alone will likely not be enough detail for someone to reproduce my results. Watching the provided YouTube videos will fill in the gaps.

Basic files of a Sparta Simulation

Input Script – The input script for a SPARTA simulation is the main input file that describes the simulation you will be running. It describes the domain, the mesh, the gas, the initial and boundary conditions, the time step size and number, the types of calculations and outputs. Inside the input scripts there are sections where the command that is used reads information from another file. These sections will read in information from files on gas composition (species), surface geometry, and collision models into the simulation when called for. Once this form and syntax that is used for these files is understood, this is a nice feature because it allows you to make changes to these parameters without changing your main input script.

Species File – This is the file that contains all the information about the gas species that will be used in the simulation. This file can contain more species than you plan to use in the simulation because when you define the gas in the input script it will only pull the information for the species with the species ID that you call for. The information stored in the species file are all the parameters for each species such as molecular weight, molecular mass, degrees of freedom, vibrational parameters, species weight and charge. Many of these parameters can be found in appendix A of [1]. Others are related to each project and if they will need to be defined for the simulation that you are running. The format of the species file is important, and it is crucial that a value is entered for every parameter even if that value is zero. SPARTA will throw an error if the expected number of entries is not present. See the species example scripts in appendices of this report and the SPARTA manual [2].

Collision file – This is the file where the collision parameters are defined for the simulation. In this file the parameters for the species involved in collisions are specified. Which are the diameter, omega (viscosity index), reference temperature, and alpha (scattering coefficient). Most of this information can also be found in appendix A of [1]. If you have a single species in your simulation, you will only have one line of information. This is the collision parameter of that species with itself. If you have multiple species, you will have a line for each species with itself and then there will be lines with the parameters for collisions between all possible collision pair combinations. For example, if the simulation gas is a mixture of Argon and Helium you will have three collision models that must be described. First, an Argon molecule colliding with another Argon molecule. Second, a Helium molecule colliding with another Helium Molecule. Finally, an Argon molecule colliding with a Helium molecule. See the example scripts that are included in the appendices of this report.

Surface file – This is the file that will describe the geometry of the surface that is in your simulation. The use of this aspect is dependent on whether you have a surface inside of the simulation domain or not. The box walls defined in the input script can be used as a surface if that is the type of simulation that you are running. Later in the report we will see an example of this when we perform the simulations from Chapter 12 of [1]. For other more complex problems a surface must be defined. In 2-D this is done by defining a list of points and a list of lines made by connecting these points. One thing to keep in mind is that when computing results SPARTA will average over the length of each line. Therefore, even though a rectangle can be defined with 4 points and 4 lines it will be necessary to discretize the sections of the surface that are of interest. A curved geometry will be made up of enough small lines to represent the curve in a reasonable manner. There are several SPARTA example files that show how to represent different shapes, scale them and even move them during the simulation.

Bird Chapter 12.2 Example from Reference [1]

The first logical step when starting to learn DSMC and SPARTA is to perform simple examples. Chapter 12 of [1] contains many 1-D examples that validate that DSMC handles the different physical phenomena correctly. To start with, the first 1-D example comes from section 12.2 of Bird's text and uses his DSMC1 program to determine the viscosity coefficient of argon. It does this with a 1-D simulation of Argon in Plane Couette flow where one surface is stationary and the other surface moves at a constant velocity. This simulation uses the Variable Hard Sphere (VHS) molecular model. I performed this simulation both with the MATLAB script I created from Bird's Fortran code, and the SPARTA simulation. I used the following parameters for both simulations. Bird states that the gas can either be initialized at 300 m/s or stationary. I tried both initial velocities and stationary appeared to give better results. There was some slight variation with the geometry and grid between the DSMC1 program and the SPARTA program. This is because Sparta can only do Quasi 1-D. All boxes are 3-D you just make the unused directions one cell within size. See the SPARTA input script in the appendix for more details.

Table 1 - Bird Section 12.2 Simulation Parameters for DSMC1 and SPARTA

Dimensions	Quasi 1-D
Surface (Wall) Spacing	1 m
Lower Surface (Wall) Velocity	0 m/s
Upper Surface (Wall) Velocity	300 m/s

Surface Temperatures (Constant)	273 K
Initial Gas Temperature	273 K
Initial Gas Velocity	0.0 m/s
Gas	Argon
Number Density	1.4e20
Collision Model	VHS (See Appendix for Details)
Mean Free Path (λ)	0.00925
Knudsen Number (Kn)	0.00925
Time Step	2.5e-6
Transient Time Steps	40,000 (DSMC1) and 50,000 (SPARTA)
Steady State (Data Collection) Time Steps	110,000 (DSMC1) and 100,000 (SPARTA)
Number of Grid Cells	200 (DSMC1 & SPARTA)
Number of Simulation Particles	10,000 (DSMC1 & SPARTA)
Number of Particles Per Cell	50 for Both DSMC1 and SPARTA

As can be seen from figures 1-5 and table 2 below. The results of my MATLAB DSMC1 version, and the SPARTA simulation matched the results from Bird section 12.2 well.

Table 2 - Bird Section 12.2 Simulation Results for Wall Velocity of 300 m/s

Variable	Bird Section 12.2	DSMC1	SPARTA
Avg Shear Stress (N/m ²)	0.00624	0.00629	0.00624
Velocity Gradient (m/s)/m	293.00	293.44	293.44
Coeff of Viscosity (N*s/m ²)	2.13e-5	2.15e-5	2.12e-5
Average Temp (K)	283.00	284.370	283.45

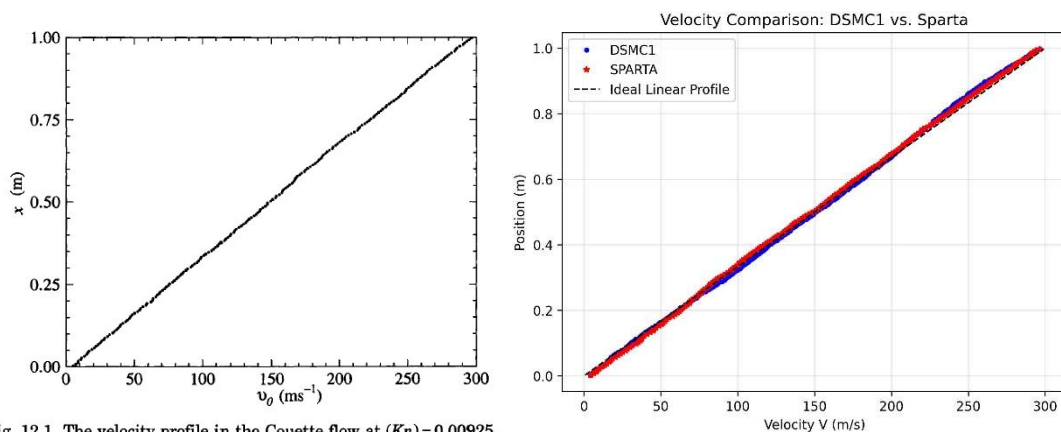


Fig. 12.1 The velocity profile in the Couette flow at $(Kn)=0.00925$.

Figure 1 - Bird Section 12.2 Velocity Profile Comparison (Bird Fig 12.1 vs. DSMC1 and SPARTA)

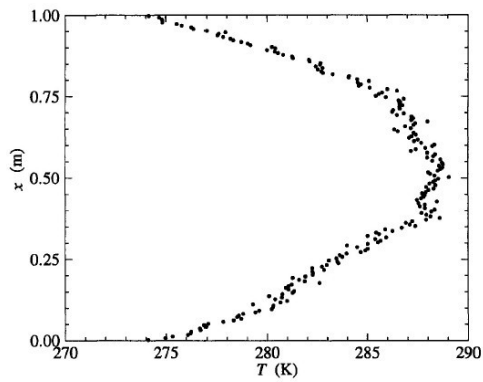


Fig. 12.2 The temperature profile in the Couette flow.

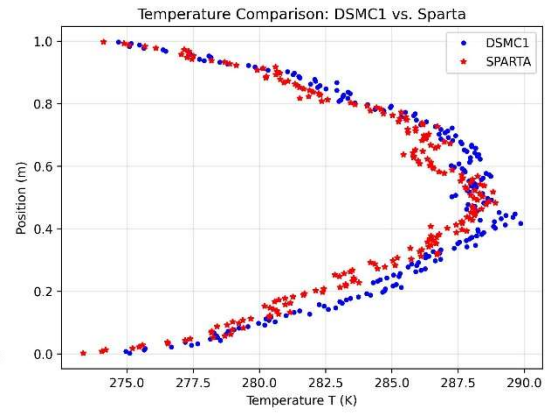


Figure 2 - Bird Section 12.2 Temperature Profile Comparison (Bird Fig 12.2 vs DSMC1 and SPARTA)

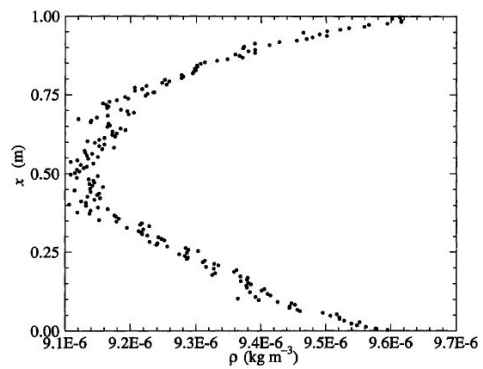


Fig. 12.3 The density profile in the Couette flow.

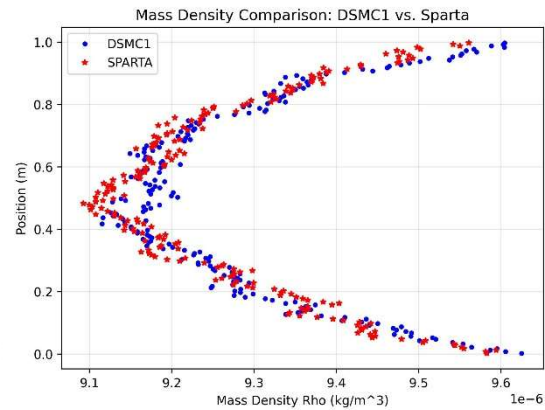


Figure 3 - Bird Section 12.2 Density Profile Comparison (Bird Fig 12.3 vs DSMC1 and SPARTA)

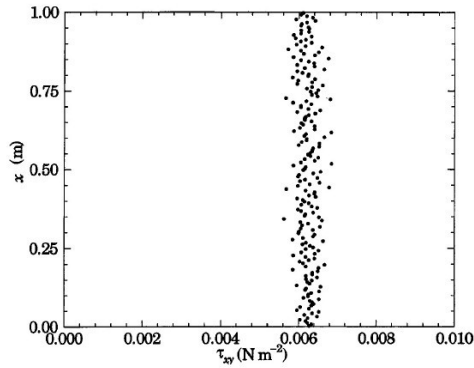


Fig. 12.4 The shear stress profile in the Couette flow.

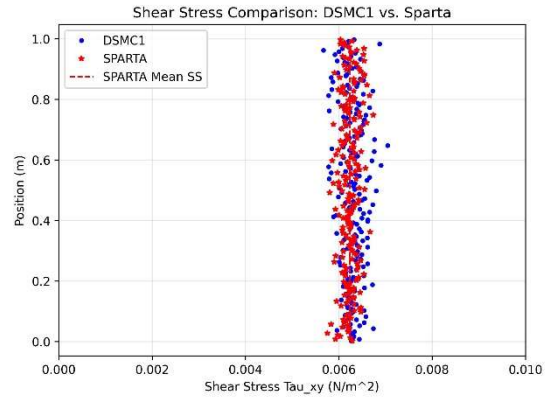


Figure 4 - Bird Section 12.2 Shear Stress Profile Comparison (Bird Fig 12.4 vs DSMC1 and SPARTA)

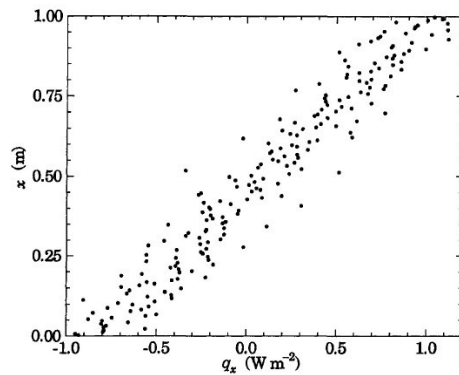


Fig. 12.5 The heat flux vector in the Couette flow.

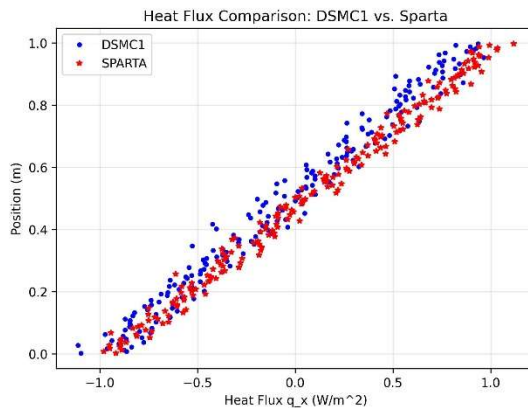


Figure 5 - Bird Section 12.2 Heat Flux Vector Comparison (Bird Fig 12.5 vs DSMC1 and SPARTA)

As can be seen in figure 5 above there is a lot of statistical scatter in the results for the Heat Flux Vector. Bird discusses this and states that “accurate results are more easily obtained for large disturbance flows than for small disturbance flows “[1]. Bird then repeats the simulation changing the wall speed to 1000 m/s which he states has a Mach number of 2.93. I also changed the wall speed and reperformed the simulation. The results can be seen below in figures 6 and 7. While there is much more nuance included in section 12.2; we have shown that a SPARTA simulation can be used for the purpose of estimating the coefficient of viscosity and replicating the results of this section. All other parameters other than wall speed are the same as in the low speed simulation.

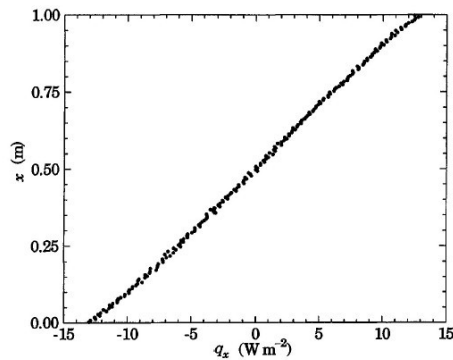


Fig. 12.6 The heat flux in the high-speed Couette flow.

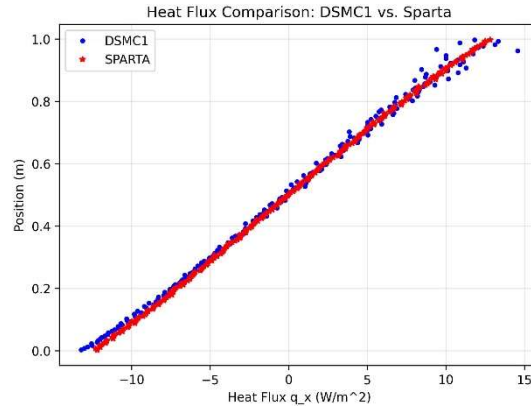


Figure 6 - Bird Section 12.2 Heat Flux Vector High Speed Comparison (Bird Fig 12.6 vs DSMC1 and SPARTA)

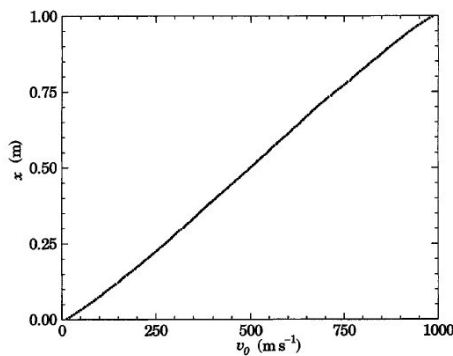


Fig. 12.7 The velocity profile in the high-speed Couette flow.

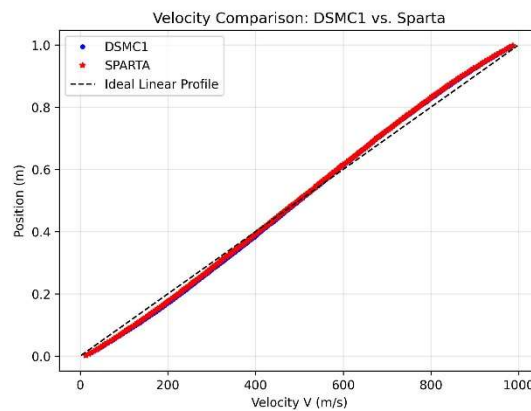


Figure 7 - Bird Section 12.2 Velocity Profile High Speed Comparison (Bird Fig 12.5 vs DSMC1 and SPARTA)

Bird Chapter 12.3 Example from Reference [1]

Next, I looked at section 12.3 of Bird's Text [1]. This section calculates the coefficient of viscosity of different mixtures of Argon and Helium at 293 K. This adds a layer of complexity to the simulation because now there are two different types of particles so there are three different particle to particle collisions possible. I used the MATLAB DSMC1 program on one of the mixtures and it was less than 0.5 percent difference from Bird's results. Running this simulation in MATLAB took approximately 4 hours on an Intel Core Ultra 7 with 32gb of ram. So, while I proved that my MATLAB translation worked again, this was the last time I ran simulations from the original Bird translations. I now focused purely on SPARTA to see if I could match Bird's results. Table 3 Below show the results of the SPARTA simulation compared to Table 12.1 of [1]. One note is that as the Helium percentage increases the amount of time to reach steady state also seems to increase in the SPARTA simulation. For

all the SPARTA simulations in the table below I ran the program with a time step of 2.5×10^{-5} seconds. I ran 250,000-time steps before I started collecting data and then ran 100,000 timesteps where a running average was collected. The only real outlier was the run when the helium percentage was 63.4 % which had a percentage difference from the measured value of 7.30 percent. When running the simulation there seemed to be large oscillations around the measured value that Bird reported. I ran a simulation again extending both the time before sampling to 350,000-time steps and then the sampling time steps to 200,000. I also adjusted how often SPARTA was sampling. With these tweaks to the simulation, I was able get a percent difference value of 1.52 %. This is similar to the difference that Bird achieved of 1.99 % with his DSMC1.

Table 3 - Initial Sparta Results for Coeff of Ar-He Mixtures Compared to Bird and Measurements

Bird Table 12.1 and SPARTA Results Coeff of Viscosity ($N \cdot S \cdot m^{-2}$)						
Helium %	Table 12.1 Measured [1]	Table 12.1 Calculated [1]	SPARTA Calculated	Bird % Difference	SPARTA % Difference	SPARTA Steps To SS / Sample
0.0	2.2275×10^{-5}	2.22×10^{-5}	2.211×10^{-5}	0.33 %	0.74 %	50k / 150k
19.9	2.2707×10^{-5}	2.24×10^{-5}	2.230×10^{-5}	1.35 %	1.79 %	50k / 150k
37.1	6346	2.31×10^{-5}	2.204×10^{-5}	0.02 %	4.57 %	50k / 150k
63.4	2.3161×10^{-5}	2.27×10^{-5}	2.218×10^{-5}	1.99 %	4.24 %	50k / 150k
100.0	1.9604×10^{-5}	1.99×10^{-5}	1.881×10^{-5}	1.53%	4.05 %	50k / 150k

I could not get the same quality of simulations as Bird did. I spent some time trying to figure out why my results were further from the measured results. One thing that I noticed is it seemed that there was a very large under shoot at the beginning of the sampling period. I ran a simulation of 37.1 % Helium and took the average values for every 1000-time steps. I noticed that there were large oscillations around the measured value. I plotted these results and they can be seen in figure 8 below. I then ran the simulations again this time slightly adjusted to try and avoid the initial undershoot.

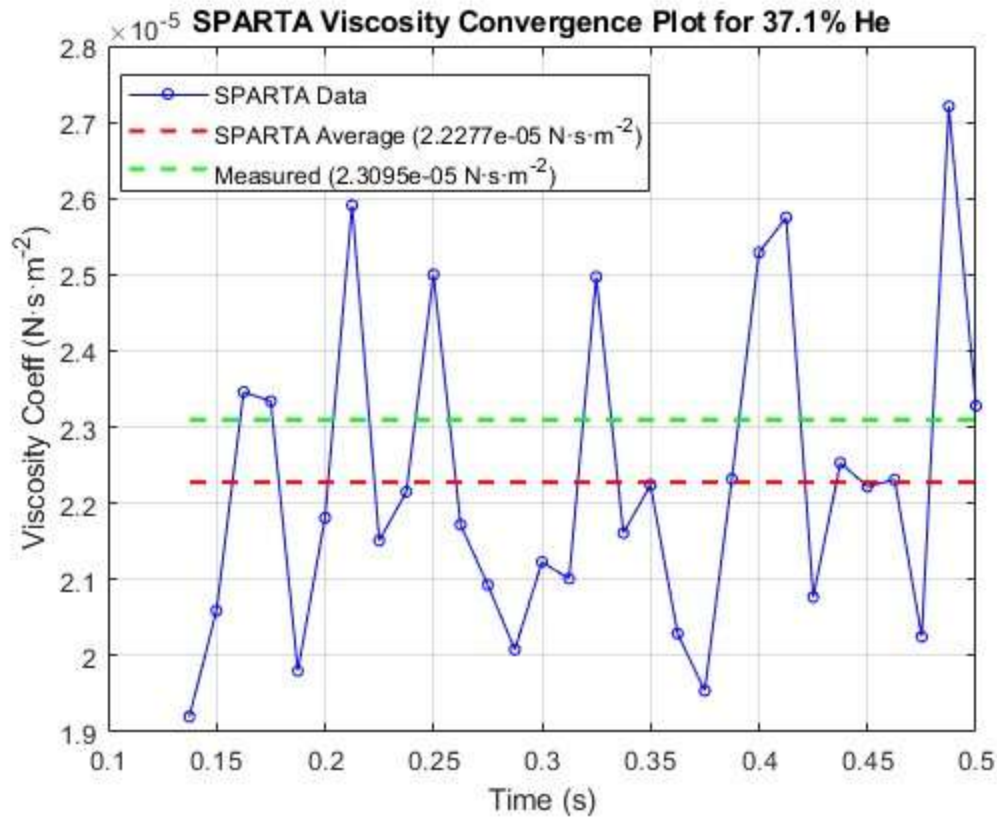


Figure 8 - 37.1 % He Viscosity Coefficient Per 1000 Time Steps

Table 4 - Adjusted Sparta Results for Coeff of Ar-He Mixtures Compared to Bird and Measurements

Bird Table 12.1 and SPARTA Results Coeff of Viscosity (N*S*m ⁻²)						
Helium %	Table 12.1 Measured [1]	Table 12.1 Calculated [1]	SPARTA Calculated	Bird % Difference	SPARTA % Difference	SPARTA Steps To SS / Sample
0.0	2.2275e-5	2.22e-5	2.256e-5	0.33 %	1.28 %	80k / 160k
19.9	2.2707e-5	2.24e-5	2.236e-5	1.35 %	1.53 %	80k / 160k
37.1	2.3095e-5	2.31e-5	2.297e-5	0.02 %	0.54 %	80k / 160k
63.4	2.3161e-5	2.27e-5	2.210e-5	1.99 %	4.58 %	80k / 160k
100.0	1.9604e-5	1.99e-5	1.826e-5	1.53%	5.58 %	80k / 160k

While the longer run times showed some improvement with some of the mixture simulations it showed a larger deviation for others. My general feeling is that there is very little value added. I have seen that it appears easier to get a reasonable estimate when the percentage of Helium is lower. I am continuing to learn more as I go along. If I were to

estimate these coefficients for an engineering problem, I would run a longer simulation and gather large number of data points and not just a running average like what was done in figure 8 above. This would allow me to visually approximate the coefficient that is somewhere between the statistical scatter that is present. Bird mentions this in his writing in section 12.3 of [1]. I would like to explore this problem more, but I must move forward with the project.

Journal Flat Plate Case from Reference [3]

The next goal for my project was to find an example journal article and recreate their SPARTA results. I found a journal article by Angelos Klothakis and Ioannis K. Nikolos titled *Comprehensive Evaluation of the Massively Parallel Direct Simulation Monte Carlo Kernel “Stochastic Parallel Rarefied-Gas Time-Accurate Analyzer” in Rarefied Hypersonic Flows* [3]. In this article they use SPARTA to simulate several cases that have been used over time to benchmark DSMC solvers and then they compare their results to other solvers and experimental results. In their article they have laid out the major parameters that they used for these calculations, but the details are sparse on the implementation. My goal for this part of the project was to use SPARTA to run the simulation and compare it to the published results of their SPARTA simulations and then provide more details into the setup of the SPARTA files for this simulation. The given problem parameters in [3] are shown in table 5 below.

Table 5 - Flat Plate Simulation Parameters from [3]

Dimensions	Quasi 2D		
Parameter	Value	Parameter	Value
Knudsen Number (Kn)	0.016	Number Density (nrho)	3.716e20
Free Stream Mean Path (λ)	0.16 mm	Surface Wall Temp	290 K
Plate Length (L)	100 mm	Surface Collisions Type	Diffuse
Plate Thickness (Tk)	5 mm	Grid Size (Rectangular)	0.5 mm X 0.5 mm
MW of N2	28.01	X Domain Size	180 mm
VHS Molecular Diameter	4.17e-10 m	Y Domain Size	205 mm
Viscosity Index (ω)	0.74	Number of Cells	147600
Reference Temp	273 K	Time Step Size	3.102e-7 s
Rotational DOF	2	Particles Per Cell Initial	10
Free Stream Velocity	1504 m/s	Time Steps to S.S.	8000
Free Stream Temp	13.32 K	Sampling Time Steps	4000

For this problem the author said that it started in a vacuum and then the flow entered and stabilized. Given the number density, the size of the simulation domain, the initial desired particles, and the number of cells, f_{num} (the number of particles represented by a simulation particle) was able to be calculated as shown below. One key is that SPARTA just assumes the size of the z direction is one when performing a 2D simulation.

$$f_{num} = \frac{n_{rho} * Volume}{particals\ per\ cell * number\ of\ cells} = \frac{3.716e20 * (0.180 * 0.205 * 1.0)}{10 * 147600}$$

$$f_{num} = 9.29e12$$

Another key is that the author said that the simulation will take rotational energy transfer into account for the collisions. Because of this I added two parameters to my collision model one was the rotational degrees of freedom. Which for diatomic nitrogen is 2, and the rotational relation which I used 0.2. I am not clear on the rotational relation yet, but this was the value that SPARTA had for all the examples where N_2 was present in the collision models. The figures that are in the journal are lacking definition in some cases, but my SPARTA simulation appears to match their results well. It is harder to see in the contour plots, but the results for the surface values track very close. My results appear to have a rougher appearance which I believe is statistical scatter. When I run the simulation much longer it smooths out considerably. I do not know if they used some post processing tool to produce their plots, or if I am missing some SPARTA command that would dampen out the scatter.

The figures below compare my SPARTA run using the time steps shown in table 5 above to the figures provided in [3]. Which ran the simulation out to 12,000 time steps. I threw out the data from the first 4,000 time steps and then took a running average of the last 8,000 time steps. For the pressure, shear stress and heat flux plots I have plotted the first and last points in the data set and then every tenth data point after that. For the interior data points, I used a 5 point central average. Which would mean for example the tenth point and the two data points on each side of it. This knocked out a lot of what I believe to be the statistical scatter and allows a good look at what the actual value is that SPARTA is predicting. Also, there is a strong singularity at the leading edge which would be expected. If the first data point was extraneous, I moved down the data set until the asymptotic nature diminished.

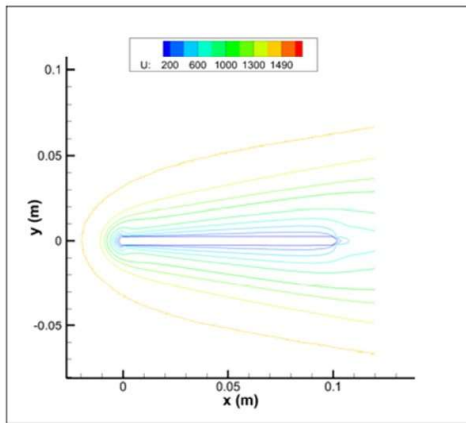


Figure 3. Velocity contours.

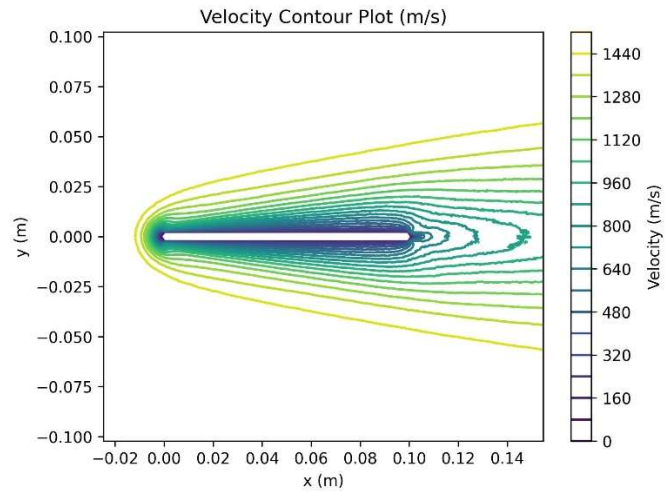


Figure 9 Klothakis & Nikolos [3] Velocity Contour vs. SPARTA Results

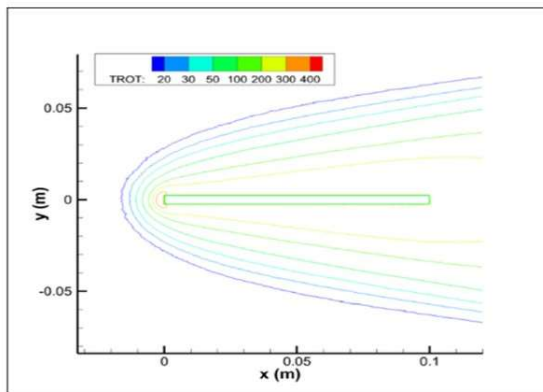


Figure 4. Rotational temperature contours.

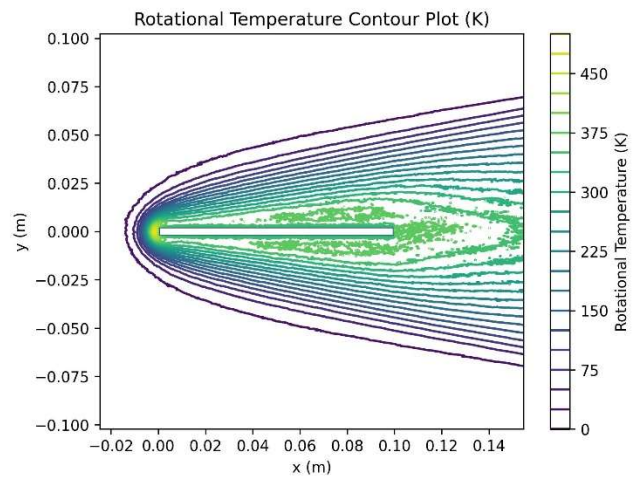


Figure 10 - Klothakis & Nikolos [3] Rotational Temp Contour vs. SPARTA Results

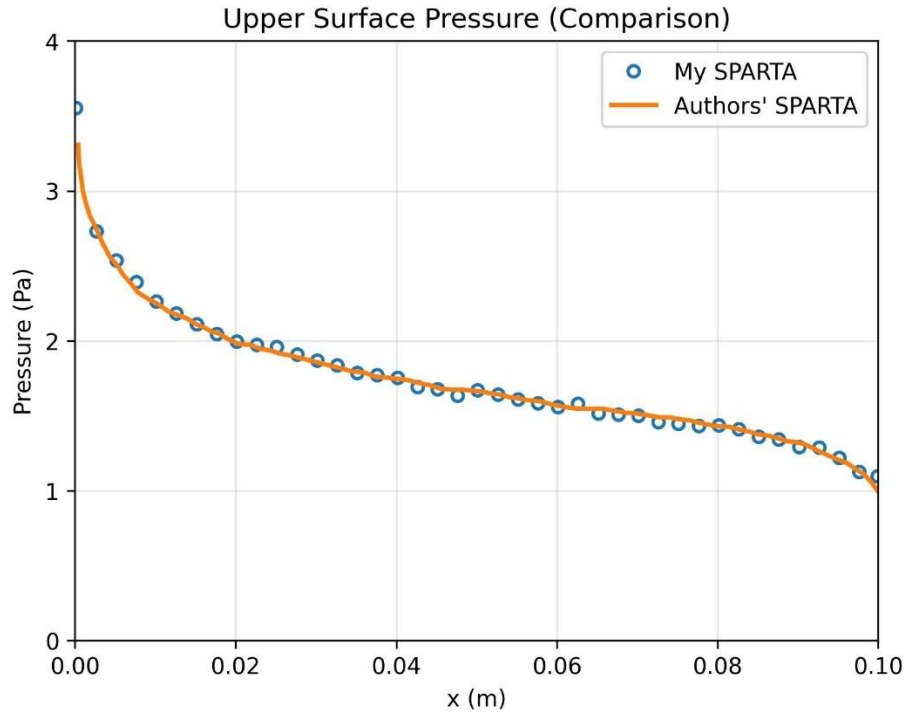


Figure 11 - Klothakis & Nikolos [3] Pressure on Upper Surface vs. SPARTA Results

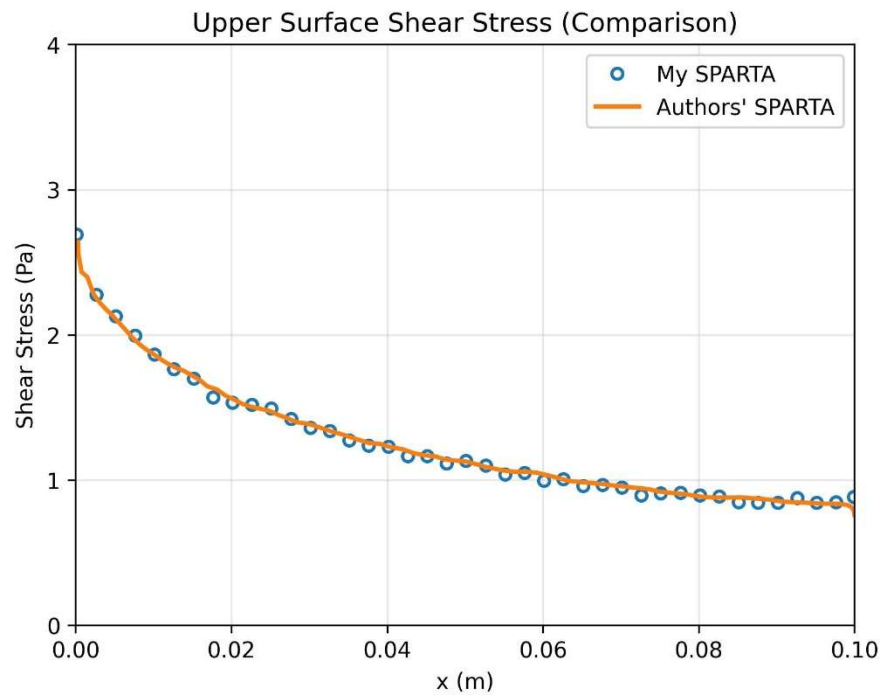


Figure 12 - Klothakis & Nikolos [3] Shear Stress on Upper Surface vs. SPARTA Results

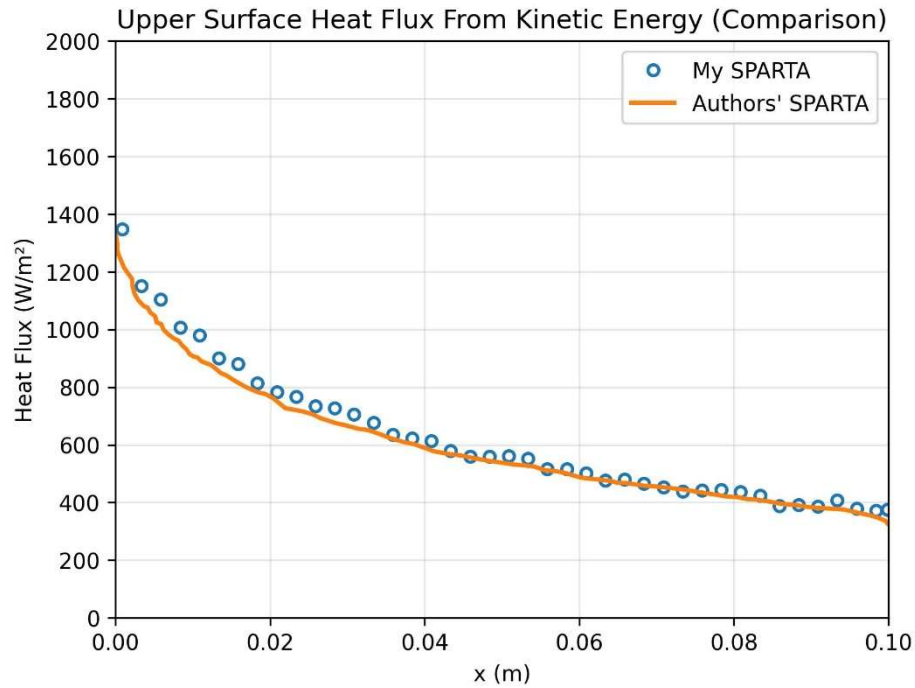


Figure 13 - Klothakis & Nikolos [3] Heat Flux on Upper Surface vs. SPARTA Energy Flux Results

I then ran the simulation out to 100,000 time steps. I threw out the data from the first 10,000 time steps and then took a running average of the last 90,000 time steps. The results are much smoother as shown below. For the pressure, shear stress and heat flux plots I have plotted I used the same smoothing technique I used previously.

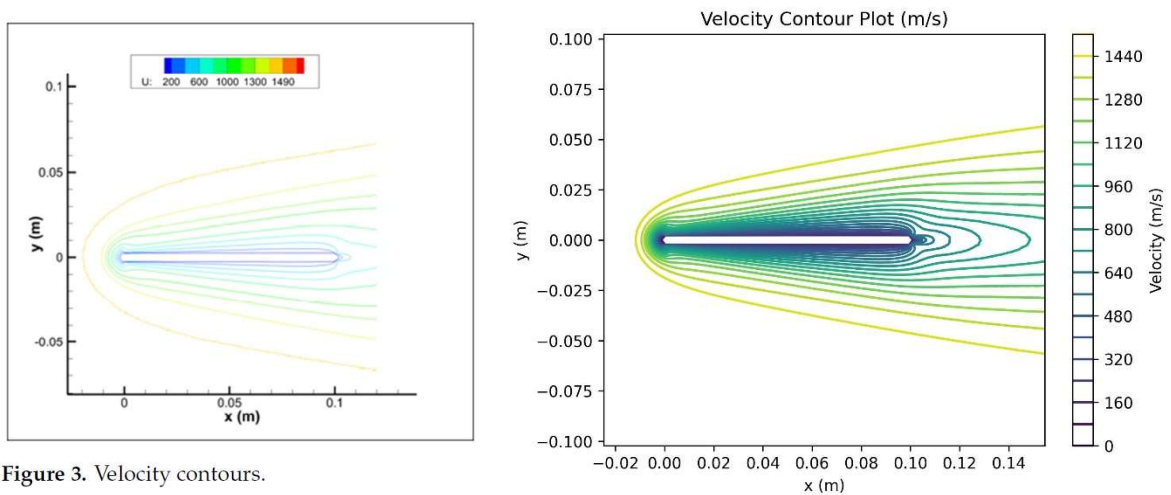


Figure 3. Velocity contours.

Figure 14 Klothakis & Nikolos [3] Velocity Contour vs. SPARTA Extended Run Results

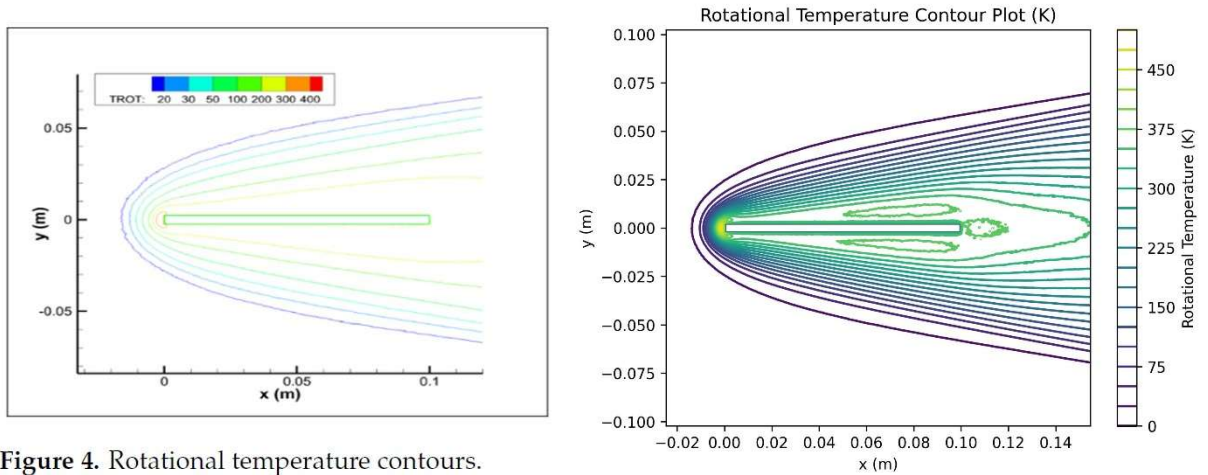


Figure 4. Rotational temperature contours.

Figure 15 - Klothakis & Nikolos [3] Rotational Temp Contour vs. SPARTA Extended Run Results

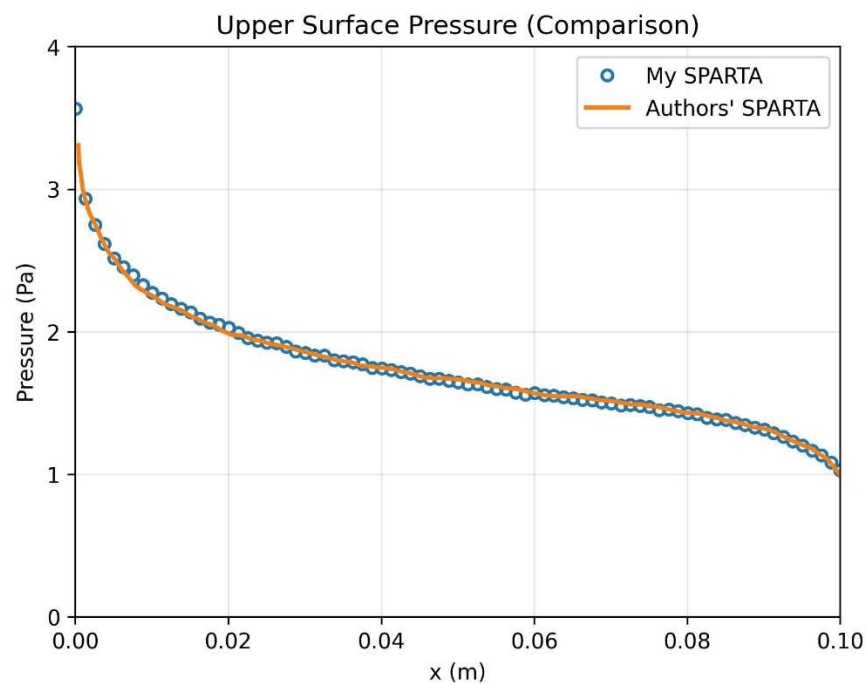


Figure 16 - Klothakis & Nikolos [3] Pressure on Upper Surface vs. SPARTA Extended Run Results

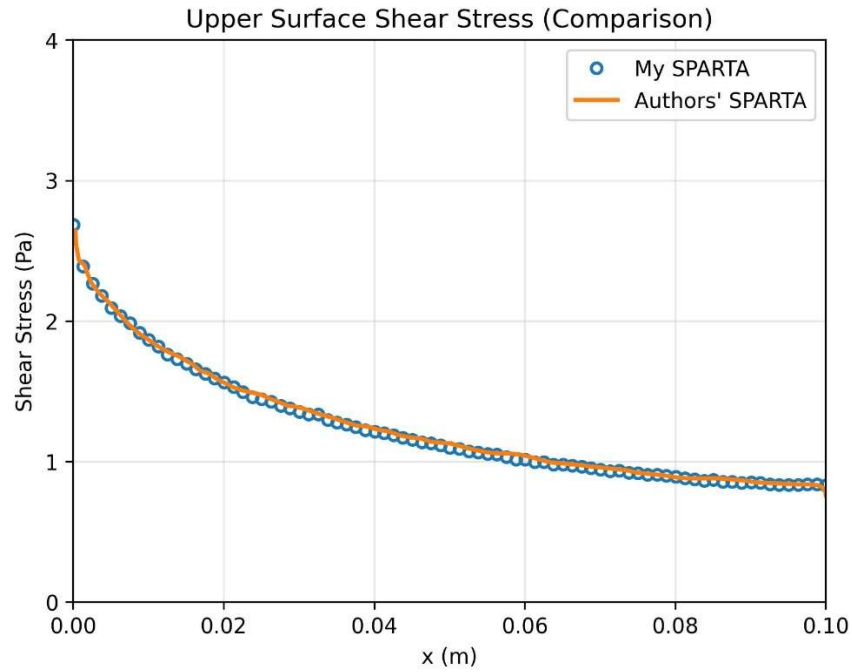


Figure 17 - Klothakis & Nikolos [3] Shear Stress on Upper Surface vs. SPARTA Extended Run Results

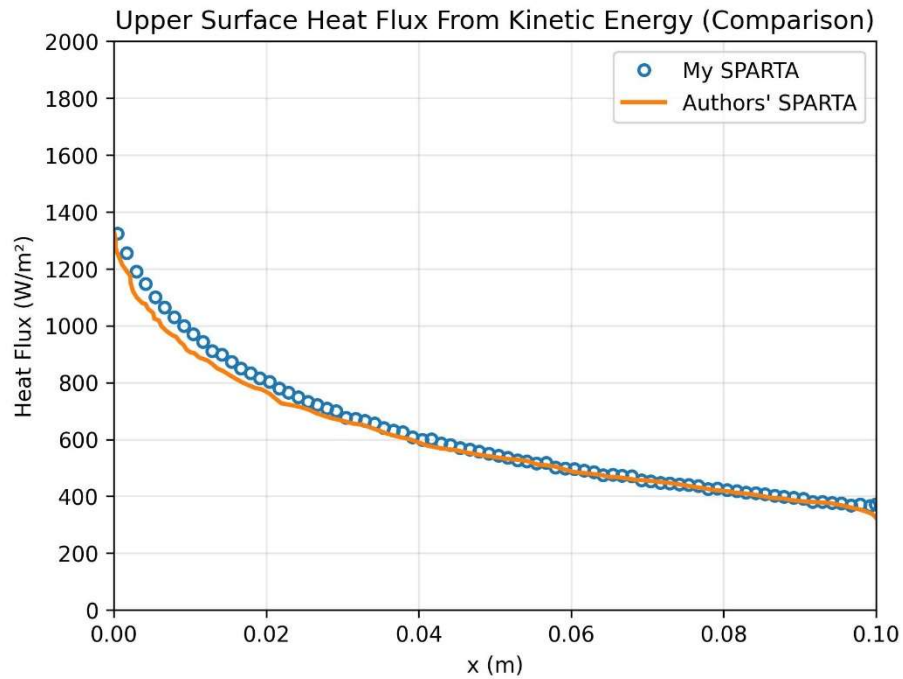


Figure 18 - Klothakis & Nikolos [3] Heat Flux on Upper Surface vs. SPARTA Extended Run Energy Flux Results

The pressure and the shear stress results match almost perfectly with the authors' results, the heat flux matches well over the last two thirds of the domain. It deviates a little bit from

the authors' reported results near the front of the plate. I am not sure why this is. There are so many different variables in SPARTA that can be adjusted I have not yet found the reason for the difference. All in all, this was a successful simulation.

Journal Interplanetary Probe Case from Reference [3]

The second simulation that Angelos Klothakis and Ioannis K. Nikolos performed in their article was the flow around an interplanetary probe. They simulated this problem in 2 ways in their article and compared their results to wind tunnel tests that had been done previously. The first method that they used was to perform a 2D axisymmetric simulation and the second was a full 3D simulation. For this project I will be attempting to replicate the axisymmetric simulation. One of the challenges with this simulation was that the surface file was much more complex than the previous flat plate example.

For the surface file I needed to sketch the surface up and then find a way to get x and y coordinates along the sketch. This is necessary because, as discussed before, SPARTA requires a list of points and list of lines to create a 2D surface. For this I downloaded *FreeCad* for Linux and within a couple of hours I was able to draw and dimension the sketch. It appears to me that the sketch in the article has some dimensions missing which would make the recreation of the surface clearer. Because of that I feel that I have a very slight difference between the authors' sketch and mine. This variation is at the front of the probe where nose radius changes. I believe this has a very small effect on the results I obtained. My axisymmetric sketch of the interplanetary probe can be seen below.

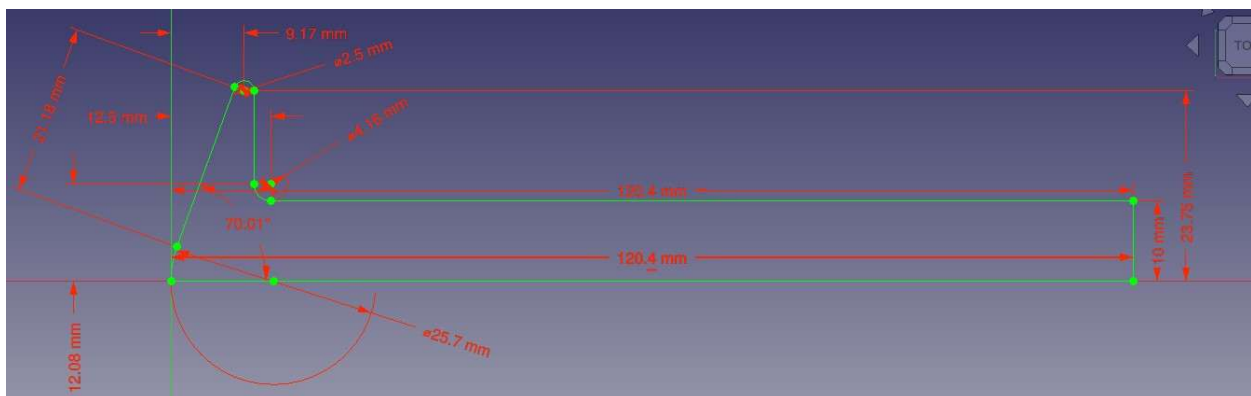


Figure 19 - Interplanetary Probe Sketch (Axisymmetric)

Once the sketch was completed, I needed to get the values for the points that will be used in my SPARTA surface file. This could be done by hand or a program written. I feel like I could write a program if needed, but I used *Claude AI* to help me write a Macro inside of *FreeCad* which would create a list of points. Inside the Macro I can change how many points to represent curves by and how many points to represent lines by. This was very helpful and saved a lot of time. The experimental values that the authors' compared their simulation to were measured with thermocouples at designated locations along the surface. Discretizing the surface well will allow me to select an area very close to those designated points in my simulation.

The information for the simulation in the article was very sparse. I had to use what was given and then make choices for the best way to run the simulation in SPARTA. The parameters that were given in [3] are in table 6 below.

Table 6 - Interplanetary Probe Flow Condition 1 Simulation Parameters from [3]

Vx (m/s)	1463.27 m/s
Free Stream Temp (T _{inf})	13.32 K
n _{rho} (Number Density)	3.869e20 m ⁻³
Wall/Surface (Probe)	300 K
Time Step (Delta t)	1.3e-07 s
F _{num}	7.2e16
Axial Domain Size	0.150 m
Radial Domain Size	0.120 m
Number of Grid Cells Axial	800
Number of Grid Cells Radial	800
Total Number of Grid Cells	640,000
Total Simulation Particles	20e6
Average Particles Per Cell	31.25
Time Steps Transient (No Data Taken)	100,000
Time Steps Steady State (Data Taken)	10,000

The one parameter that seemed to be the challenge for me was the number of particles in the simulation. There are two ways to weight the cells for particle distribution for an axisymmetric problem in SPARTA. You can either weight the cells based on cell volume or based on radial distance from the axis of symmetry. The example that SPARTA has in the download used radial weighting, so I started with that approach. The simulation ran and gave somewhat reasonable results, but the results were better in some areas than others. This simulation took approximately 13 hours to run on my laptop. With this set up I had approximately 3 million particles in the simulation, which is much less than the 20 million the author reported. This gave me an average of approximately 4.5 particles per cell. This

was much less than is desirable and much less than the authors reported using. I next tried volume based weighting. When I attempted to run the simulation, it appeared to not be running at all. I changed the stats output to show me how the simulation progressed every time step. This showed that the simulation was progressing, but so slowly that it may take a month to complete. The number of particles was over 60 million after only a few dozen timesteps.

I tried several ways to control the number of particles in the simulation. The way I settled on achieving this control was to use the radial weighting and reduce Fnum incrementally. I divided Fnum by two and ran the simulation again. This gave me between 6 and 7 million particles in simulation and took more than a day to run. The results were slightly improved, so I divided Fnum by two again and started the simulation again. This gave me approximately 13 million particles in the simulation, and it could not really be determined if the results were better. The results at the nose were slightly worse while areas down the length were better. This simulation took about three days to run. I plan to try one more time and see the numbers and results and time, but this will likely take a week or more. I believe that this problem has allowed me to understand the limit of what can be reasonably done without a strong cluster of CPU cores to use. I am sure that using the program with the virtual box is also causing the simulation to run slower than it would on a dedicated Linux machine.

I also tried both VHS and VSS collision models. The authors' did not state which model they used in their simulation. I performed most of the simulations with the VSS models, and it gave me better results at least for the 6.5 million particle simulation.

One note is that as I would expect, the number density at the front of the probe near the stagnation point is much higher than around the rest of the body. I feel this is leading me to have results that are reasonably close to the experimental results and the authors' results at the front of the probe. I believe that with the number density being the highest at the front of the probe, the simulation particles are also denser in that area which allows proper collision rate estimations. This area of high density can be seen in figure 26 below.

In the documented results below, I am using a VSS collision model and they come from the first change of Fnum that I incorporated to increase the particle count in the simulation. This simulation contained approximately 6.5 million simulation particles with radial weighting, and a Fnum of $3.6e16$. The simulation took between one and two days to complete.

The results in the forward part of the cone are relatively close to the experimental measurements, while there is a large difference between results and the experimental data

towards the back of the probe. The authors discuss in [3] how in the experiment there was likely trouble in the measuring of the flow data towards the back of the cone due to the level of rarefaction. This could mean that the simulation results towards the back of the probe are closer to the true value than the wind tunnel measurements. The heat flux at the front of the cone is also several orders of magnitude greater than at the rear. This would be the primary area of interest in an engineering application. I also have some deviations with the shape of my surface and with respect to where on the surface my values are being taken. While these deviations are relatively small this can still affect the results.

There are still several unknowns for this simulation that I have not been able to determine. What commands and scheme did they use for the weighting of the particles in the cells? What type of collision model did they use, and where did the parameters for that model come from? What energy exchange modes did they include in their simulation? I performed all my simulations with rotational energy exchange turned on and vibrational turned off. I did try once with the vibrational energy exchange on and the results were much further off than with out it included.

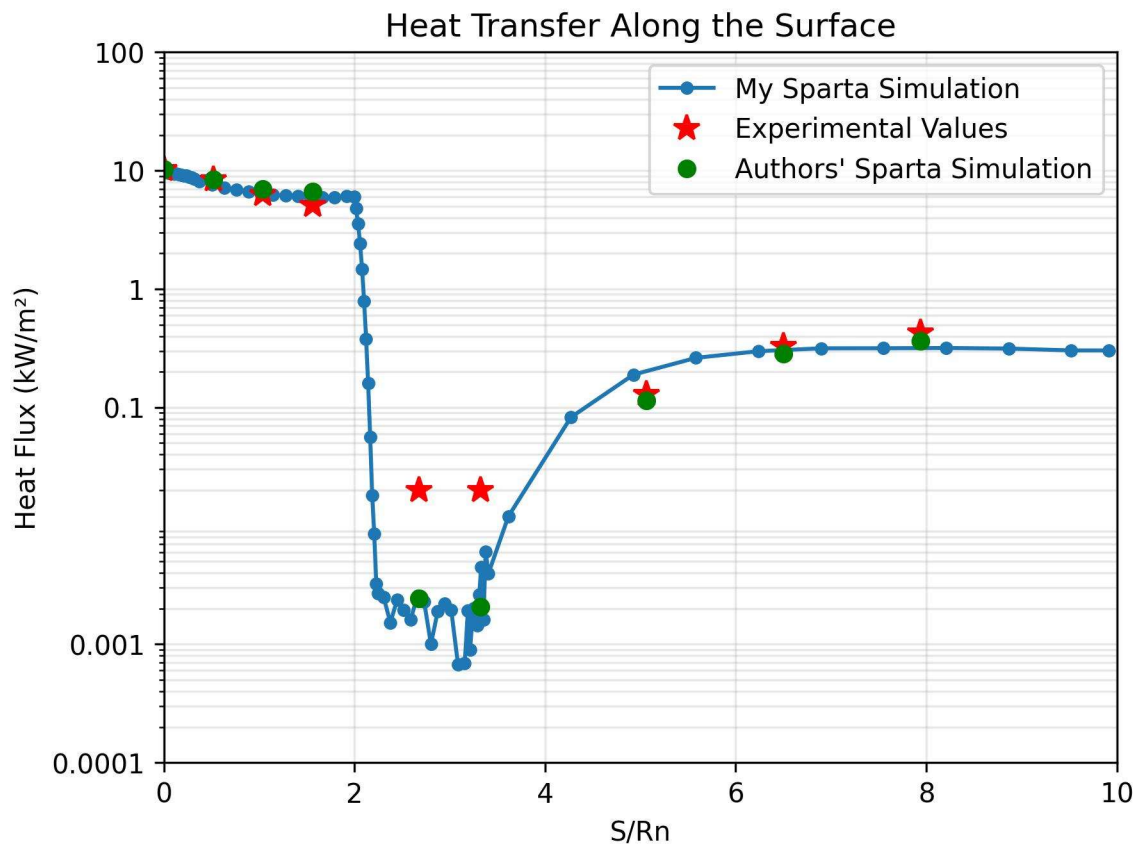


Figure 20 - Heat Flux Comparison (Data from [3] Included)

Matched Points Table

Evaluation points	Closest S_Rn points	My Values	Experimental values	My_Exp %diff	Journ Values	My_Jour % Diff	Jour_Exp % Diff
0.0	0.0	9.821	10.23	4.0	10.228	4.0	0.0
0.52	0.53	7.466	8.39	11.0	8.393	11.0	0.0
1.04	1.045	6.392	6.33	1.0	6.979	8.4	10.3
1.56	1.56	6.008	5.12	17.3	6.627	9.4	29.4
2.68	2.675	0.003	<0.02	0.0	0.002	23.0	0.0
3.32	3.322	0.004	<0.02	0.0	0.002	108.8	0.0
5.06	5.059	0.207	0.13	59.2	0.115	79.9	11.5
6.5	6.502	0.31	0.33	6.0	0.285	9.0	13.7
7.94	7.945	0.318	0.43	26.2	0.363	12.6	15.6

Table 7 - Comparison of the Heat Flux at the Thermocouple Locations (Data from Klothakis & Nikolos [3] Included)

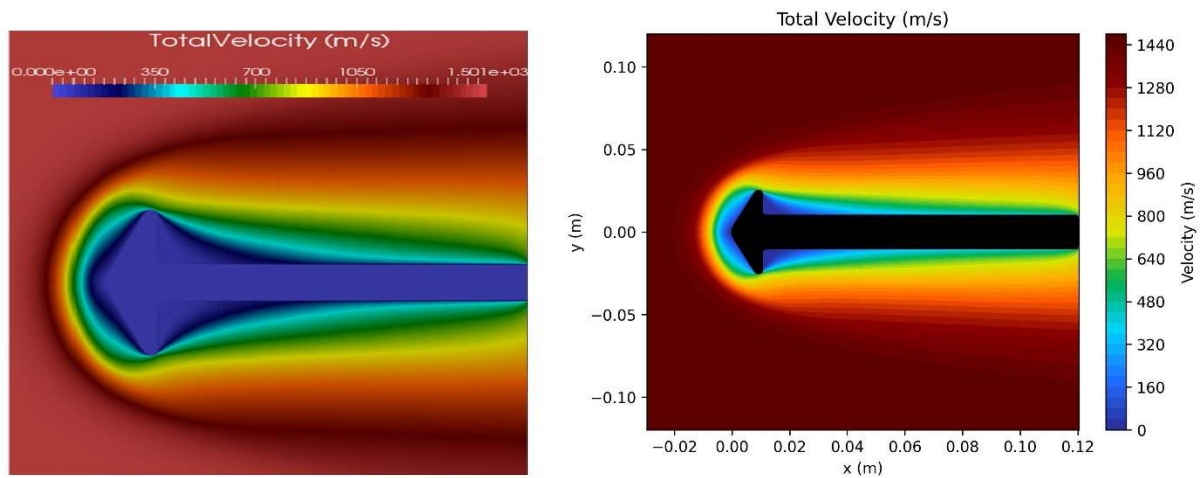


Figure 21 - Klothakis & Nikolos [3] Total Velocity Contour vs. SPARTA Run Results

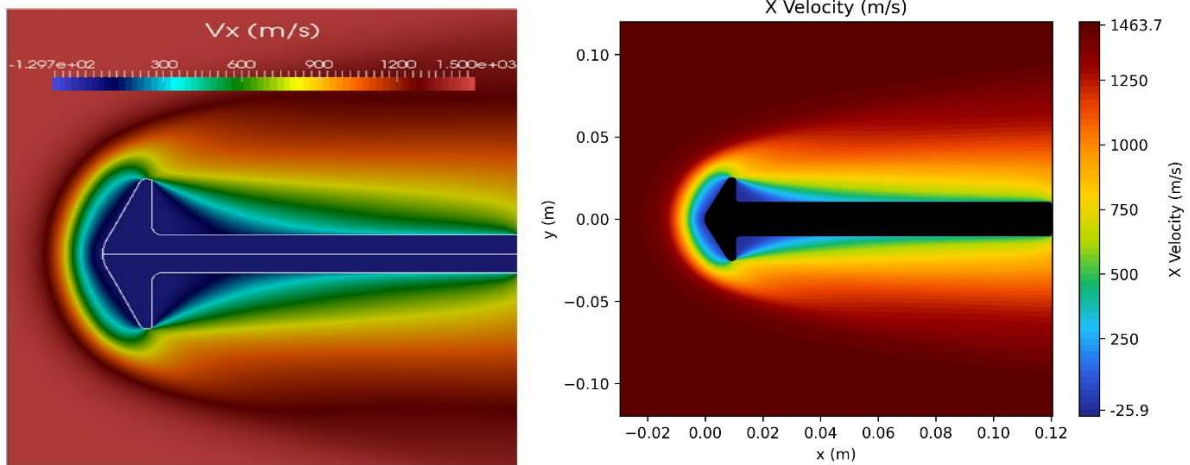


Figure 22 - Klothakis & Nikolos [3] X-Velocity (Axial) Contour vs. SPARTA Run Results

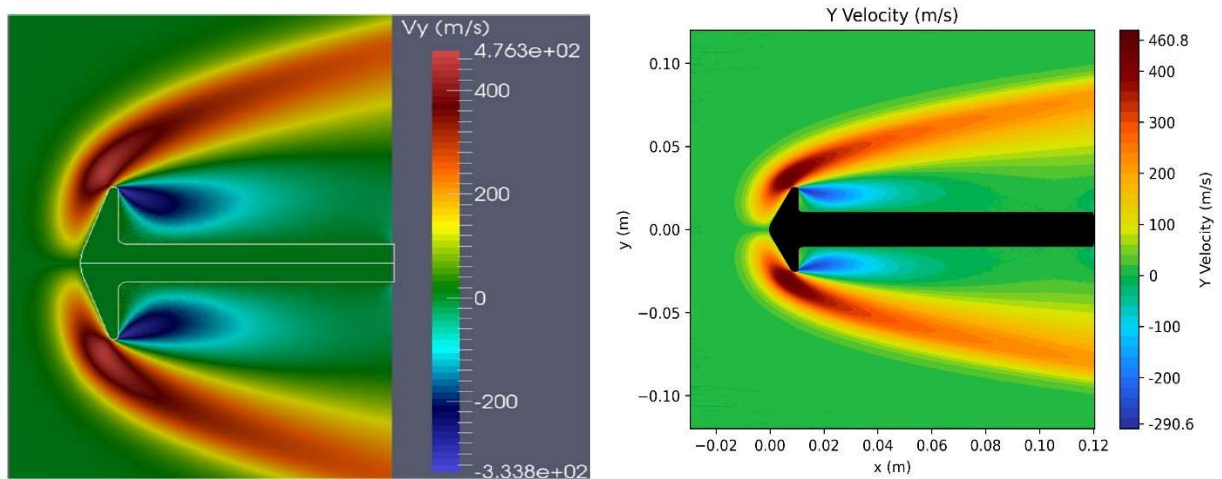


Figure 23 - Klothakis & Nikolos [3] Y-Velocity (Radial) Contour vs. SPARTA Run Results

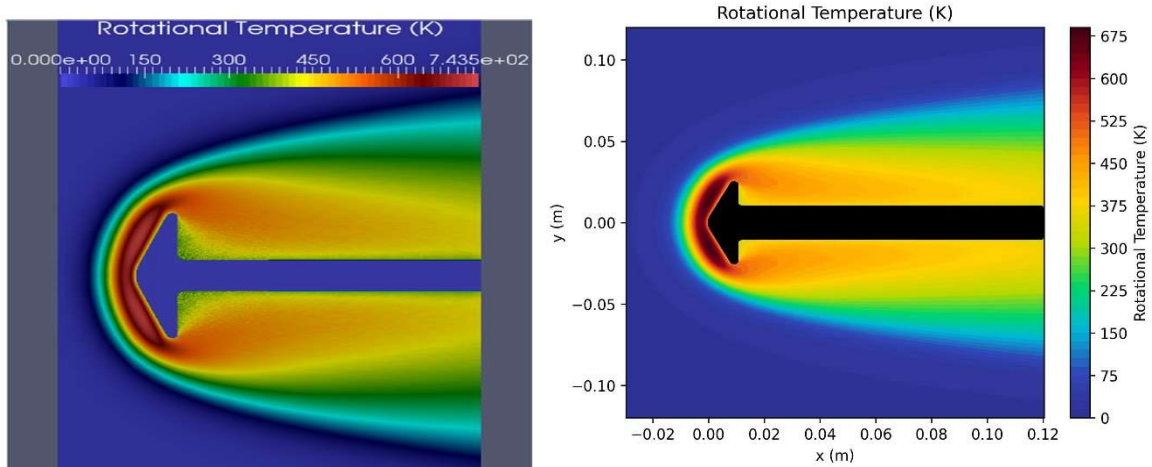


Figure 24 - Klothakis & Nikolos [3] Rotational Temperature Contour vs. SPARTA Run Results

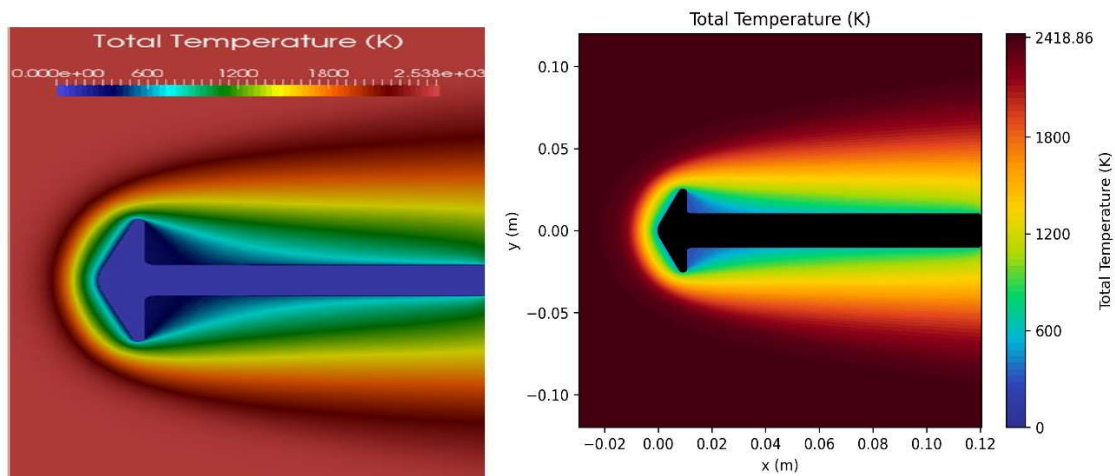


Figure 25 - Klothakis & Nikolos [3] Total Temperature Contour vs. SPARTA Run Results

There is reasonably close agreement for the total temperature between my simulation and the authors' reported results. My max temperature is slightly lower which agrees with my slightly lower values for the maximum heat flux. Inside the affected flow we have similar results which have the shape that one would expect. You can see the line in my plot where the shock resided by looking at the large temperature changes. Its location and form are as I would expect them to be.

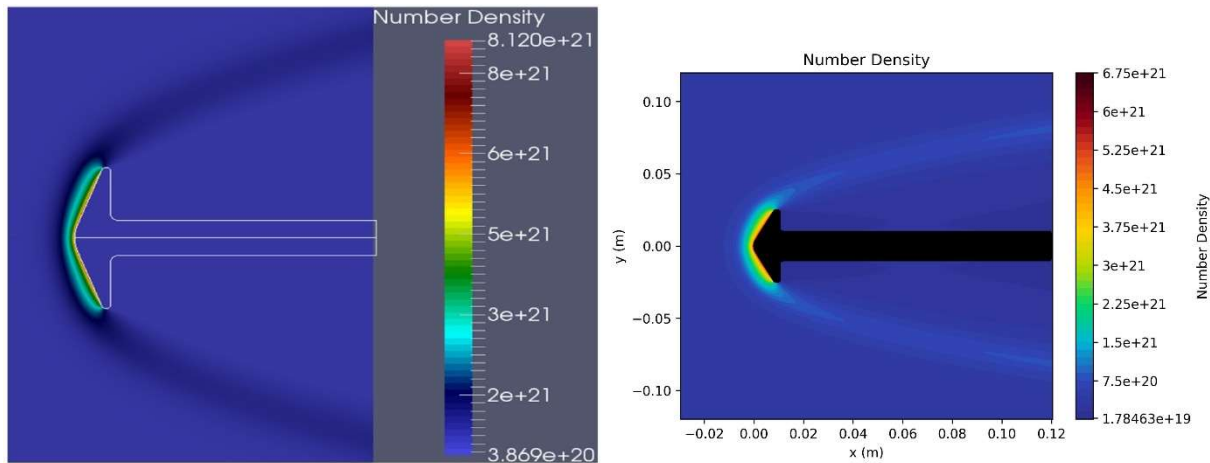


Figure 26 - Klothakis & Nikolos [3] Number Density Contour vs. SPARTA Run Results

Conclusion

My goal for this project was to expand my knowledge and learn something new in a self-study manner. There were many new challenges that I encountered during this project. All the programs that I used for this project were open source programs. The experience I gained with Linux and these open source programs will allow me to progress in exploring this subject and other subjects in the future. My current job does not require this kind of work and so the commercial CFD, FEA and coding programs are not available to me. This has been a great opportunity to start getting comfortable learning how to use many new tools.

I also wanted to embrace AI tools that can be used in engineering. During this project I learned what these tools are good at and what they are not. I never felt that AI was doing something that I could not do, but I felt that in a matter of seconds with my guidance it could do something that would take me hours to perform. I have learned that using AI is a skill and especially that what AI produces needs to be verified.

I have gained some familiarity with rarified gas dynamics, DSMC and SPARTA. When this project idea was presented to me by Dr. Gonzales, I had to research what SPARTA and DSMC were. After this project I now feel like I have a basic understanding of the uses, process and challenges. It is also clear to me that I have just scratched the surface of the subject and there is so much more to learn. I completed and investigated a total of four problems. All had different aspects to them that made them unique. I reproduced Bird's one dimension codes into MATLAB and applied them to the first two problems. I also have

created a GitHub repository and a YouTube channel where the details and files for the SPARTA simulations I performed can be found. While there is a learning curve to SPARTA, I feel I am now able to perform simulations of increased complexity. However, to do so my knowledge of the subject needs to increase and I would need access to a stronger processing cluster to perform the simulations.

I am very thankful Dr. Gonzales recommended this project to me, and for the time I was able to spend learning from and getting to know him. I also would like to thank Dr. Qiao for his willingness to step in to advise me on the completion of this project.

GitHub Repository Link

<https://github.com/Awall005/ABW-SPARTA-EXAMPLE-FILES>

YouTube Channel Link

<https://www.youtube.com/@AaronWallaceMECHENG>

Works Cited

- [1] G. A. Bird, *Molecular gas dynamics and the direct simulation of gas flows*. Oxford: Clarendon Press ; New York, 1994.
- [2] “SPARTA Documentation,” *Github.io*, 2026. <https://sparta.github.io/doc/Manual.html> (accessed Apr. 02, 2026).
- [3] A. Klothakis and I. K. Nikolos, “Comprehensive Evaluation of the Massively Parallel Direct Simulation Monte Carlo Kernel ‘Stochastic Parallel Rarefied-Gas Time-Accurate Analyzer’ in Rarefied Hypersonic Flows—Part A: Fundamentals,” *Computation*, vol. 12, no. 10, p. 198, Oct. 2024, doi: <https://doi.org/10.3390/computation12100198>.

Appendix A – SPARTA Files for Bird 12.2 Problem

Species File (ar.species)

```
# Species data

# ID
# Molwt (amu) - Google This - I rounded slightly because this is what
sparta examples appeared to do.
# Molmass (kg) - Table A1
# Rotational dof - Typically, 0 = monatomic species, 2 = diatomic, 3 =
anything else.
# RotRel - Not Used in this Simulation
# Vibrational dof - Not Used in this Simulation
# VibRel - Not Used in this Simulation
# VibTemp (K) - Not Used in this Simulation
# species wt - Used to weight simulation vs real particals. This can be 1
and then do it in the input script
# charge - Not Used in this Simulation

# Note that all the listed rotational and vibrational values must be
specified for each species,
# but in cases where they are not used by SPARTA, they can simply be
specified as 0.0

Ar  40.00  6.63E-26  0    0.0  0    0.0  0.0      1.0    0.0
```

Collision Model File (ar.vhs)

```
# VHS collision model parameters for each species

# species (or species pair)
# diameter (m) - From Table A3 Bird for VSS, From Table A2 if VHS
# omega - From Table A1 Bird (This is the Same for VHS and VSS)
# tref - From Table A2 or A3 Bird
# alpha - From Table A3 Bird (VSS Table A3, If VHS Model alpha = 1)

Ar    4.17E-10    0.81    273.15    1.0
```

SPARTA Input File (in.couette_bird_12.2)

```
#####
#####
# SPARTA Couette Flow - Bird Chapter 12.2 Argon Viscosity Measurement
# FULL BIRD SIMULATION: Matches DSMC1.FOR exactly
#
# Expected:  $\mu = 2.117 \times 10^{-5}$  Pa·s at 273K (Bird Section 12.2)
#
```

```

# FULL BIRD SIMULATION PARAMETERS:
#   Phase 1: 50,000 steps transient (discarded) = 0.125 seconds
#   Phase 2: 100,000 steps steady state (saved) = 0.25 seconds
#   Total: 150,000 steps = 0.375 seconds
#####

#####

# INITIALIZATION
#####

seed                12345
dimension           2
global              gridcut 0.0 comm/sort yes

#####

# BOUNDARY CONDITIONS
#####

boundary            p s p
surf_collide        1 diffuse 273.0 1.0
surf_collide        2 diffuse 273.0 1.0 translate 300.0 0 0
bound_modify        ylo collide 1
bound_modify        yhi collide 2

#####

# DOMAIN GEOMETRY
#####

create_box          0 0.01 0 1.0 -0.0005 0.0005

#####

# GRID SETUP
#####

create_grid         1 200 1
balance_grid        rcb cell

#####

# GAS SPECIES AND MIXTURE
#####

species             ar.species Ar
mixture             gas Ar temp 273.0 vstream 0.0 0.0 0.0

```



```

#####
#####
# PARTICLE WEIGHTING AND INITIALIZATION
#####
#####

global          fnum 1.4e14 nrho 1.4e20
create_particles gas n 10000

#####
#####
# COLLISION MODEL
#####
#####

collide          vss gas ar.vhs

#####
#####
# COMPUTE: GRID PROPERTIES
#####
#####

compute          gcomp grid all all u v w massrho nrho temp
compute          gtcomp thermal/grid all all temp
compute          gpcomp thermal/grid all all press
compute          hfcomp eflux/grid all species heatx heaty heatz
compute          mfcomp pflux/grid all species momxy

#####
#####
# TIME-AVERAGING FIXES (will be reset after transient)
#####
#####

fix              gfix ave/grid all 1 500 500 c_gcomp[*]
fix              gtfix ave/grid all 1 500 500 c_gtcomp[*]
fix              gpfix ave/grid all 1 500 500 c_gpcomp[*]
fix              hfix ave/grid all 1 500 500 c_hfcomp[*]
fix              mfix ave/grid all 1 500 500 c_mfcomp[*]

#####
#####
# OUTPUT FILES (will be enabled after transient)
#####
#####

dump             gdump grid all 500 grid.*.dat id yc f_gfix[*]
f_gtfix[*] f_gpfix[*] f_hfix[*] f_mfix[*]

#####
#####
# STATISTICS OUTPUT

```

```

#####
#####

stats          500
stats_style    step cpu np nattempt ncoll nscoll nscheck

#####
#####
# TIME INTEGRATION
#####
#####

timestep       2.5e-6

#####
#####
# PHASE 1: TRANSIENT PERIOD (DISCARD DATA)
#####
#####

undump         gdump

print
"=====
print          "FULL BIRD CHAPTER 12.2 SIMULATION"
print
"=====
print          "PHASE 1: Running transient period (50000 steps)"
print          "          No output files will be generated"
print          "          Simulation time: 0.125 seconds"
print
"=====

run            50000

#####
#####
# PHASE 2: STEADY STATE PERIOD (SAVE DATA)
#####
#####

print
"=====
print          "PHASE 2: Resetting statistics for steady state"
print
"=====

unfix          gfix
unfix          gtfix
unfix          gpfix
unfix          hfix
unfix          mfix

fix            gfix ave/grid all 1 500 500 c_gcomp[*] ave running

```

```

fix          gtfix ave/grid all 1 500 500 c_gtcomp[*] ave running
fix          gpfix ave/grid all 1 500 500 c_gpcomp[*] ave running
fix          hfix ave/grid all 1 500 500 c_hfcomp[*] ave running
fix          mfix ave/grid all 1 500 500 c_mfcomp[*] ave running

dump          gdump grid all 50000 grid.*.dat id yc f_gfix[*]
f_gtfix[*] f_gpfix[*] f_hfix[*] f_mfix[*]

print          "          Running steady state (100000 steps)"
print          "          Simulation time: 0.25 seconds"
print
"=====

run          100000

#####
#####
# SIMULATION COMPLETE
#####
#####

print
"=====
print          "SIMULATION COMPLETE - FULL BIRD CHAPTER 12.2"
print
"=====
print          "Total timesteps: 150000 (matching Bird DSMC1.FOR
exactly)"
print          "Total simulation time: 0.375 seconds"
print          "Transient discarded: 50000 steps (0.125 s)"
print          "Steady state saved: 100000 steps (0.25 s)"

```

Appendix B – SPARTA Files for Bird 12.3 Problem

Species File (gas.species)

```
# Species data

# ID
# Molwt (amu) - Google This
# Molmass (kg) - Table A1
# Rotational dof - Typically, 0 = monatomic species, 2 = diatomic, 3 =
anything else.
# RotRel
# Vibrational dof
# VibRel
# VibTemp (K)
# species wt - Used to weight simulation vs real particals. This can be 1
and then do it in the input script
# charge

# Note that all the listed rotational and vibrational values must be
specified for each species, but in cases where they
# are not used by SPARTA, they can simply be specified as 0.0.

Ar  40.00  66.3E-27  0    0.0  0    0.0  0.0  1  0.0
He  4.00   6.65E-27  0    0.0  0    0.0  0.0  1  0.0
```

Collision Model File (arhe.vss)

```
# VSS collision model parameters for each species

# species (or species pair)
# diameter (m) - (From Table A3 Bird for VSS, From T.A2 if VHS) Avg if
Species Pair
# omega - From Table A1 Bird (This is the Same for VHS and VSS)
# tref - From Table A3 Bird
# alpha - From Table A3 Bird (VSS Table A3, If VHS Model alpha = 1)
# for species pair collisions get omega and alpha from Bird table A4

Ar    4.11E-10    0.81 273    1.40
He    2.30E-10    0.66 273    1.26

Ar He   3.205E-10 0.725 273    1.64
```

SPARTA Input File (in.couette_bird_12.3_37.1Percent_He)

```
#####
#####
# SPARTA Couette Flow - Bird Chapter 12.3 Argon/Helium Mixture Viscosity
Measurement
#####
#####

#####
#####
# INITIALIZATION
#####
#####

seed                12345
dimension           2
global              gridcut 0.0 comm/sort yes

#####
#####
# BOUNDARY CONDITIONS
#####
#####

boundary            p s p
surf_collide        1 diffuse 293.0 1.0
surf_collide        2 diffuse 293.0 1.0 translate 300.0 0 0
bound_modify        ylo collide 1
bound_modify        yhi collide 2

#####
#####
# DOMAIN GEOMETRY
#####
#####

create_box          0 0.01 0 1.0 -0.0005 0.0005

#####
#####
# GRID SETUP
#####
#####

create_grid         1 220 1
balance_grid        rcb cell

#####
#####
# GAS SPECIES AND MIXTURE
```

```

#####
#####

species          gas.species Ar He
mixture          gas Ar He temp 293.0 vstream 300.0 0.0 0.0 nrho 1.4e20
mixture          gas Ar frac 0.629
mixture          gas He frac 0.371

#####
#####
# PARTICLE WEIGHTING AND INITIALIZATION
#####
#####

# fnum calc nrho = 1.4e20, 220 cells, 50 particals per cell, fake vol =
0.01*1*1
# fnum = 1.4e20*(0.01*1*1)/(220*50) = 1.27e14, give 11023 particals

global           fnum 1.27e14
# nrho 1.4e20
create_particles  gas n 0

#####
#####
# COLLISION MODEL
#####
#####

collide          vss gas arhe.vss

#####
#####
# COMPUTE: GRID PROPERTIES
#####
#####

compute          gcomp grid all all u v w massrho nrho temp
compute          gtcomp thermal/grid all all temp
compute          gpcomp thermal/grid all all press
compute          hfcomp eflux/grid all species heatx heaty heatz
compute          mfcomp pflux/grid all species momxy

#####
#####
# TIME-AVERAGING FIXES (will be reset after transient)
#####
#####

fix              gfix ave/grid all 10 1000 10000 c_gcomp[*]
fix              gtfix ave/grid all 10 1000 10000 c_gtcomp[*]
fix              gpfix ave/grid all 10 1000 10000 c_gpcomp[*]
fix              hfix ave/grid all 10 1000 10000 c_hfcomp[*]
fix              mfix ave/grid all 10 1000 10000 c_mfcomp[*]

```

```

#####
#####
# OUTPUT FILES (will be enabled after transient)
#####
#####

dump                      gdump grid all 500 grid.*.dat id yc f_gfix[*]
f_gtfix[*] f_gpfix[*] f_hfix[*] f_mfix[*]

#####
#####
# STATISTICS OUTPUT
#####
#####

stats                      500
stats_style                step cpu np nattempt ncoll nscoll nscheck

#####
#####
# TIME INTEGRATION
#####
#####

timestep                  2.5e-6

#####
#####
# PHASE 1: TRANSIENT PERIOD (DISCARD DATA)
#####
#####

undump                    gdump

print
"=====
print                      "BIRD CHAPTER 12.3 SIMULATION"
print
"=====
print                      "PHASE 1: Running transient period (250000 steps)"
print
"=====

run                        250000

#####
#####
# PHASE 2: STEADY STATE PERIOD (SAVE DATA)
#####
#####

print
"=====
print                      "PHASE 2: Resetting statistics for steady state"

```

```

print
"=====

unfix          gfix
unfix          gtfix
unfix          gpfix
unfix          hfix
unfix          mfix

fix            gfix ave/grid all 1 500 500 c_gcomp[*] ave running
fix            gtfix ave/grid all 1 500 500 c_gtcomp[*] ave running
fix            gpfix ave/grid all 1 500 500 c_gpcomp[*] ave running
fix            hfix ave/grid all 1 500 500 c_hfcomp[*] ave running
fix            mfix ave/grid all 1 500 500 c_mfcomp[*] ave running

dump           gdump grid all 50000 grid.*.dat id yc f_gfix[*]
f_gtfix[*] f_gpfix[*] f_hfix[*] f_mfix[*]

print          "          Running steady state (100000 steps)"
print
"=====

run            100000

#####
#####
# SIMULATION COMPLETE
#####
#####

print
"=====
print          "SIMULATION COMPLETE - BIRD CHAPTER 12.3"
print
"=====

```


Appendix C – SPARTA Files for Flat Plate Problem

Species File (N2.species)

```
# Species data

# ID
# Molwt (amu) - Google
# Molmass (kg) - Table A1 Bird
# Rotational dof - Journal Info
# RotRel
# Vibrational dof
# VibRel
# VibTemp (K)
# species wt
# charge

N2  28.01    4.65E-26  2    0.2    2    1.90114E-5    3371.0    1.0    0.0
```

Collision Model File (N2.vhs)

```
# VHS collision model parameters for each species

# diameter (m) - Table A2 Bird
# omega - Table A1 Bird
# tref - Table A2 Bird
# alpha - 1.0 for VHS

N2  4.17E-10    0.74  273  1.0
```

Surface File (data.flatplate403) –

The file to log to include all lines. First and last five points and lines are shown. Go to [github](#) for full file.

```
dummy 2d line test

403 points
403 lines

Points

1 0.00000 0.0025
2 0.00025 0.0025
3 0.00050 0.0025
```

```

4 0.00075 0.0025
5 0.00100 0.0025

```

Points missing . . .

```

399 0.09950 0.0025
400 0.09975 0.0025
401 0.10000 0.0025
402 0.00000 -0.0025
403 0.10000 -0.0025

```

Lines

```

1 1 2
2 2 3
3 3 4
4 4 5
5 5 6

```

Lines missing . . .

```

399 399 400
400 400 401
401 401 403
402 403 402
403 402 1

```

SPARTA Input File (in.flat_Plate)

```

#####
#####
# 2d flow around a flat plate
#
# Note:
# - The "comm/sort" option to the "global" command is used to match MPI
  runs.
# - The "twopass" option is used to match Kokkos runs.
# The "comm/sort" and "twopass" options should not be used for production
  runs.
#####
#####

```

```

##### INITIALIZATION #####
seed          12345
dimension      2
global         gridcut 0.0 comm/sort yes

```

```

##### BOUNDARY CONDITIONS #####

```

```

# This comes from the article's text. x is inflow and outflow, y is
# reflective. z being periodic because it is not used and should not affect
# simulation
boundary          o r p

##### DOMAIN GEOMETRY #####
create_box          -0.025 0.155 -0.1025 0.1025 -0.00025 0.00025

##### GRID SETUP #####
create_grid          360 410 1
balance_grid          rcb cell
# Rebalancing the how the processors are loaded as was done in artical
fix                  1 balance 4000 1.3 rcb cell

##### GAS SPECIES AND MIXTURE #####
species              N2.species N2
mixture              gas N2 vstream 1504.0 0 0 temp 13.34 nrho 3.716e20

##### SIMULATION PARTICAL WEIGHTING #####
# fnum was calculated based off of a desired 10 particals in each cell and
# nrho amount shown in mixture
global              fnum 9.29e12

##### GAS COLLISION MODEL #####
collide              vss gas N2.vhs
collide_modify        rotate smooth vibrate no

fix                  in emit/face gas xlo twopass

##### SURFACE GEOMETRY AND COLLISION MODEL #####
read_surf            data.flatplate403 trans 0 0 0
surf_collide          1 diffuse 290.0 1.0
surf_modify           all collide 1

##### COMPUTE: GRID PROPERTIES #####
compute              gcomp grid all all u v w massrho nrho trot
compute              gtcomp thermal/grid all all temp
compute              gpcomp thermal/grid all all press
compute              hfcomp eflux/grid all species heatx heaty heatz
compute              mfcomp pflux/grid all species momxy

##### TIME AVG FIXES GRID PROPETIES (will be reset after transient)
#####
fix                  gfix ave/grid all 10 100 1000 c_gcomp[*]
fix                  gtfix ave/grid all 10 100 1000 c_gtcomp[*]
fix                  gpfix ave/grid all 10 100 1000 c_gpcomp[*]
fix                  hfix ave/grid all 10 100 1000 c_hfcomp[*]
fix                  mfix ave/grid all 10 100 1000 c_mfcomp[*]

##### OUTPUT FILES GRID PROPERTIES (will be enabled after transient)
#####

```

```

dump                gdump grid all 1000 grid.*.dat id xc yc f_gfix[*]
f_gtfix[*] f_gpfix[*] f_hfix[*] f_mfix[*]

##### COMPUTE/FIX/DUMP SURFACE PROPERTIES #####
compute            scomp surf all gas press shx shy shz etot ke
compute            sgeom property/surf all id xc yc

fix                sfix ave/surf all 10 100 1000 c_scomp[*]

dump                sdump surf all 1000 surf.*.dat c_sgeom[*] f_sfix[*]

##### STATISTICS OUTPUT #####
stats              100
stats_style        step cpu np nattempt ncoll nscoll nscheck

##### GLOBAL TIMESTEP SIZE #####
timestep           3.102e-7

##### IMAGE DUMP IF NEEDED #####
#dump              2 image all 50 image.*.ppm type type pdiam 0.001 &
#                  surf proc 0.01 size 512 512 axes yes 0.9 0.02
#dump_modify       2 pad 4

##### TRANISIENT RUN #####
undump             gdump
undump             sdump

print
"=====
print              "FULL SUPERSONIC FLATPLATE SIMULATION"
print
"=====
print              "PHASE 1: Running transient period (8000 steps)"
print
"=====

run                8000

#### STEADY STATE PERIOD (SAVE DATA) #####

print
"=====
print              "PHASE 2: Resetting statistics for steady state"
print
"=====

unfix              gfix
unfix              gtfix

```

```

unfix          gpfix
unfix          hfix
unfix          mfix
unfix          sfix

fix            gfix ave/grid all 1 1000 1000 c_gcomp[*] ave running
fix            gtfix ave/grid all 1 1000 1000 c_gtcomp[*] ave running
fix            gpfix ave/grid all 1 1000 1000 c_gpcomp[*] ave running
fix            hfix ave/grid all 1 1000 1000 c_hfcomp[*] ave running
fix            mfix ave/grid all 1 1000 1000 c_mfcomp[*] ave running

fix            sfix ave/surf all 1 1000 1000 c_scomp[*] ave
running

dump           gdump grid all 1000 grid.*.dat id xc yc f_gfix[*]
f_gtfix[*] f_gpfix[*] f_hfix[*] f_mfix[*]

dump           sdump surf all 1000 surf.*.dat c_sgeom[*] f_sfix[*]

print          "          Running steady state (4000 steps)"
print
"=====

run            4000

#####
#####
# SIMULATION COMPLETE
#####
#####

print
"=====
print          "SIMULATION COMPLETE - FULL Run"
print
"=====

```

Appendix D – SPARTA Files for Interplanetary Probe Problem

Species File (air.species)

```
# Species data

# ID
# Molwt (amu)
# Molmass (kg)
# Rotational dof
# RotRel
# Vibrational dof
# VibRel
# VibTemp (K)
# species wt
# charge

O2  32.00      5.31E-26  2    0.2    2    5.58659E-5    2256.0    1.0
0.0
N2  28.016     4.65E-26  2    0.2    2    1.90114E-5    3371.0    1.0
0.0
O   16.00      2.65E-26  0    0.0    0    0.0           0.0       1.0
0.0
N   14.008     2.325E-26  0    0.0    0    0.0           0.0       1.0
0.0
NO  30.008     4.98E-26  2    0.2    2    7.14285E-4    2719.0    1.0
0.0
O2+ 32.00      5.31E-26  2    0.2    2    5.58659E-5    2256.0    1.0
1.0
N2+ 28.016     4.65E-26  2    0.2    2    1.90114E-5    3371.0    1.0
1.0
O+  16.00      2.65E-26  0    0.0    0    0.0           0.0       1.0
1.0
N+  14.008     2.325E-26  0    0.0    0    0.0           0.0       1.0
1.0
NO+ 30.008     4.98E-26  2    0.2    2    7.14285E-4    2719.0    1.0
1.0
e   0.001      9.10938188E-31 0 0.0    0    0.0           0.0       1.0    -
1.0
```

Collision Model File (air.vss)

```
# VSS collision model parameters for each species

# diameter (m)
# omega
# tref
# alpha

O2  3.96E-10    0.77  273.15  1.4
N2  4.07E-10    0.74  273.15  1.6
```

O	3.0E-10	0.80	273.15	1.0
N	3.0E-10	0.80	273.15	1.0
NO	4.0E-10	0.80	273.15	1.0
O2+	3.96E-10	0.77	273.15	1.4
N2+	4.07E-10	0.74	273.15	1.6
O+	3.0E-10	0.80	273.15	1.0
N+	3.0E-10	0.80	273.15	1.0
NO+	4.0E-10	0.80	273.15	1.0
e	7.0E-13	0.50	273.15	1.0

If there a mixture you will have to add info for cross species collisions.

Surface File (data.probe_fine)

This file is to large to place here. Go to - [https://github.com/Awall005/ABW-SPARTA-EXAMPLE-FILES/blob/main/Probe%20in%20N2%20\(Axisymmetric\)/data.probe_fine](https://github.com/Awall005/ABW-SPARTA-EXAMPLE-FILES/blob/main/Probe%20in%20N2%20(Axisymmetric)/data.probe_fine)

SPARTA Input File (in.probe_axisymmetric)

```
#####
#####
# 2d axisymmetric flow around a probe with diffuse reflections
#
# Note:
# - The "comm/sort" option to the "global" command is used to match MPI
runs.
# - The "twopass" option is used to match Kokkos runs.
# The "comm/sort" and "twopass" options should not be used for production
runs.
#####

##### INITIALIZATION #####
seed          12345
dimension      2
global         gridcut 0.0 comm/sort yes

##### BOUNDARY CONDITIONS #####
boundary       o ar p

##### DOMAIN GEOMETRY #####
create_box     -0.0296 0.12040 0.0 0.12 -0.5 0.5

##### GRID SETUP #####
create_grid    800 800 1
balance_grid   rcb cell
# Rebalancing the how the processors are loaded as was done in artical
fix           1 balance 40000 1.3 rcb cell
```

```

##### GAS SPECIES AND MIXTURE #####
species          air.species N2
mixture          air N2 vstream 1463.27 0.0 0.0 temp 13.32 nrho
3.869e20

##### SIMULATION PARTICAL WEIGHTING #####
# fnum given in text
global           fnum 3.6e16
#global          weight cell volume
global           weight cell radius

##### GAS COLLISION MODEL #####
collide          vss air air.vss
collide_modify   rotate smooth vibrate no

fix              in emit/face air xlo twopass

##### SURFACE GEOMETRY AND COLLISION MODEL #####
read_surf        data.probe_fine origin 0 0 0
surf_collide     1 diffuse 300.0 1.0
surf_modify      all collide 1

##### COMPUTE: GRID PROPERTIES #####
compute          gcomp grid all all u v w massrho nrho trot
compute          gtcomp thermal/grid all all temp
compute          gpcomp thermal/grid all all press
compute          hfcomp eflux/grid all species heatx heaty heatz
compute          mfcomp pflux/grid all species momxy

##### COMPUTE/FIX/DUMP SURFACE PROPERTIES ####
compute          scomp surf all air press shx shy shz etot ke
compute          sgeom property/surf all id xc yc area

##### STATISTICS OUTPUT #####
stats            100
stats_style      step cpu np nattempt ncoll nscoll nscheck

##### GLOBAL TIMESTEP SIZE #####
timestep         1.3e-7

##### TRANISENT RUN #####

print
"=====
print              "FULL Interplanetary Probe SIMULATION"
print
"=====
print              "PHASE 1: Running transient period (8000 steps)"
print
"=====

```



```
run                100000
```

```
#### STEADY STATE PERIOD (SAVE DATA) #####
```

```
print
"=====
print          "PHASE 2: Resetting statistics for steady state"
print
"=====

fix            gfix ave/grid all 1 1000 1000 c_gcomp[*] ave running
fix            gtfix ave/grid all 1 1000 1000 c_gtcomp[*] ave running
fix            gpfix ave/grid all 1 1000 1000 c_gpcomp[*] ave running
fix            hfix ave/grid all 1 1000 1000 c_hfcomp[*] ave running
fix            mfix ave/grid all 1 1000 1000 c_mfcomp[*] ave running

fix            sfix ave/surf all 1 1000 1000 c_scomp[*] ave
running

dump            gdump grid all 1000 grid.*.dat id xc yc f_gfix[*]
f_gtfix[*] f_gpfix[*] f_hfix[*] f_mfix[*]

dump            sdump surf all 1000 surf.*.dat c_sgeom[*] f_sfix[*]

print          "          Running steady state (4000 steps)"
print
"=====
```

```
run                20000
```

```
#####
#####
# SIMULATION COMPLETE
#####
#####

print
"=====
print          "SIMULATION COMPLETE - FULL Run"
print
"=====
```